

Lupus Nephritis: Proliferative Forms (WHO III, IV)

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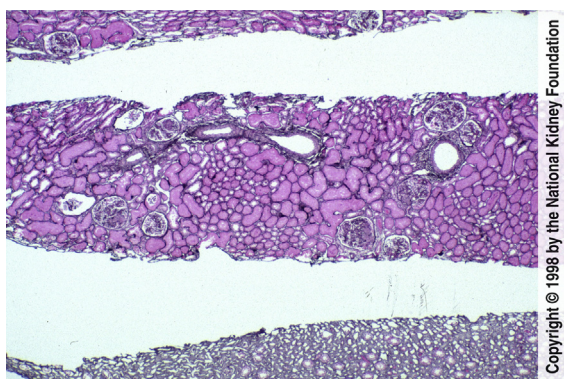


Fig 1. Proliferative lupus nephritis is defined by endocapillary proliferation due to subendothelial deposits, and may be focal (less than 50% of glomeruli involved) or diffuse (greater than 50% of glomeruli involved). The lesions are qualitatively identical in focal and diffuse proliferative lupus nephritis (WHO Class III and IV, respectively). The focal segmental nature of proliferation in WHO Class III lupus nephritis is illustrated. Some glomeruli appear to have only minor mesangial expansion, whereas others have marked segmental endocapillary proliferation and even necrosis. No tubulointerstitial nephritis or fibrosis is seen in this biopsy. There is also no sclerosis. Thus, activity is moderate, indicated by proliferation and necrosis but relatively few crescents, and chronicity is minimal, with absence of significant fibrosis or sclerosis (Jones' silver stain; original magnification $\times 100$).

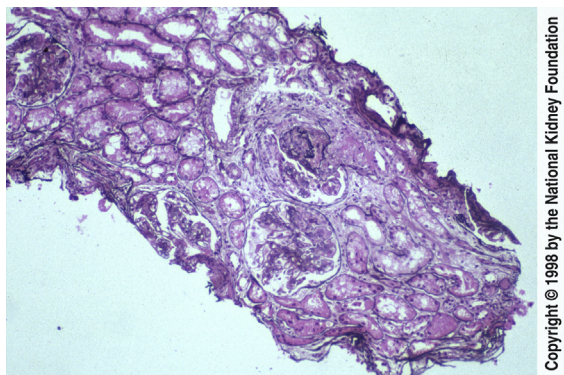


Fig 2. This biopsy shows findings typical of proliferative lupus, which in this case was diffuse (WHO Class IV). One glomerulus shows mild mesangial expansion and very segmental endocapillary proliferation, whereas marked segmental endocapillary proliferation and necrosis are present in other glomeruli. The necrosis extends to the arteriole, and a small cellular crescent has formed with the rupture of Bowman's capsule and surrounding early fibrosis and macrophages (Jones' silver stain; original magnification $\times 200$).

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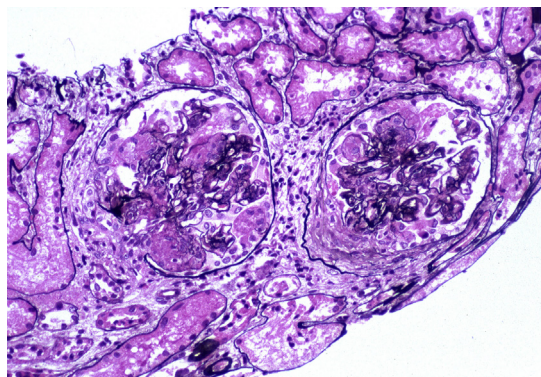
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Am J Kidney Dis. 31(6):E1-E2.2.

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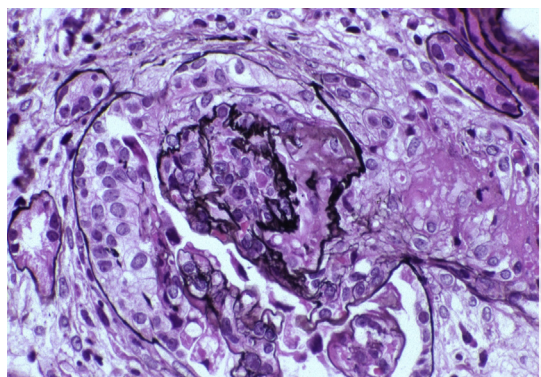
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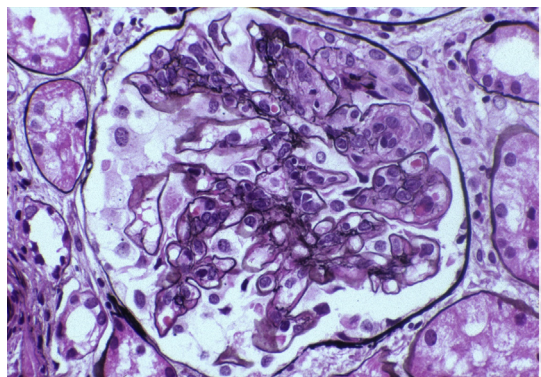
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Fig 3. The segmental necrosis and proliferation with mild increase in extracapillary cells, not reaching the level of a crescent, are illustrated in this case of proliferative lupus nephritis (the process was focal in this biopsy, WHO III, which involves less than 50% of glomeruli). There is surrounding early interstitial fibrosis and lymphoplasmocytic infiltrate with thickening, fibrosis, and rupture of Bowman's capsule in the glomerulus on the right (Jones' silver stain; original magnification $\times 200$).



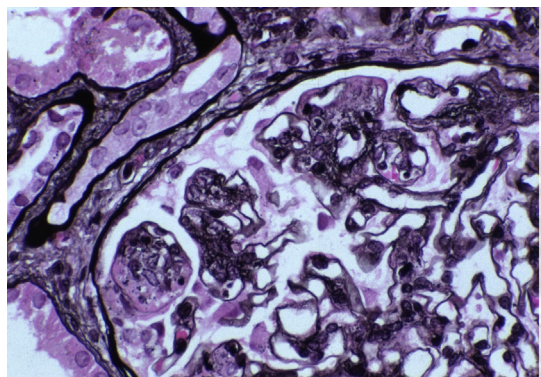
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Fig 4. Diffuse proliferative lupus nephritis with segmental endocapillary proliferation and eosinophilic subendothelial sausage-shaped deposits visualized underlying the silver-stained basement membrane segmentally. There is also an early cellular crescent and rupture of Bowman's capsule (Jones' silver stain; original magnification $\times 400$).



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Fig 5. The segmental nature of proliferation in proliferative lupus nephritis (WHO IV) is seen in this glomerulus, where half of the tuft is distorted by marked endocapillary proliferation with occasional infiltrating cells. Segmental areas of basement membrane splitting and eosinophilic subendothelial deposits and mesangial eosinophilic deposits are visualized (Jones' silver stain; original magnification $\times 400$).



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Fig 6. Extensive glomerular basement membrane alterations can be seen corresponding to the extensive deposits in this case of diffuse proliferative lupus nephritis (WHO Class IV). There is segmental glomerular basement membrane splitting with eosinophilic, large, sausage-shaped subendothelial deposits. In other areas, bubbly holes can be seen in tangential sections of the basement membrane, representing deposits that are subepithelial (deposits do not take up the silver stain and are outlined as holes in these cuts). Segmental endocapillary proliferation and mesangial proliferation are also present (Jones' silver stain; original magnification $\times 600$).

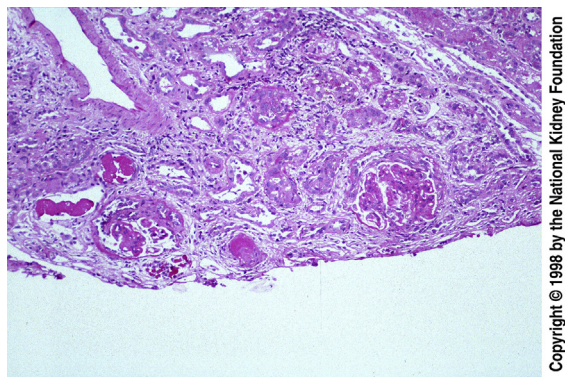


Fig 7. In this case of diffuse proliferative lupus nephritis (WHO Class IV), there is moderate chronicity, evidenced by moderate diffuse interstitial fibrosis, glomerulosclerosis, and fibrocellular crescents. There is still remaining activity of this proliferative process, evidenced by endocapillary proliferation and tubulointerstitial lymphocytic infiltrate (periodic acid-Schiff; original magnification $\times 200$).

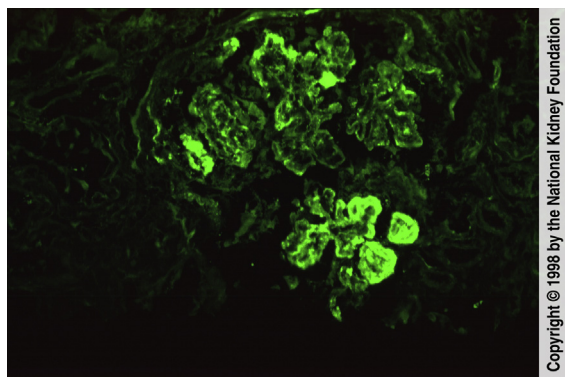


Fig 8. Segmental capillary wall and mesangial deposits in proliferative lupus nephritis are shown. In this case, proliferation involved more than half of the glomeruli, ie, diffuse proliferative lupus nephritis (WHO Class IV). Some of the larger capillary wall deposits are sausage-shaped with smooth outer contours, indicating they are subendothelial in location. Other smaller granular deposits along the capillary wall may correspond to smaller subepithelial deposits (polyvalent antisera, immunofluorescence; original magnification $\times 200$).

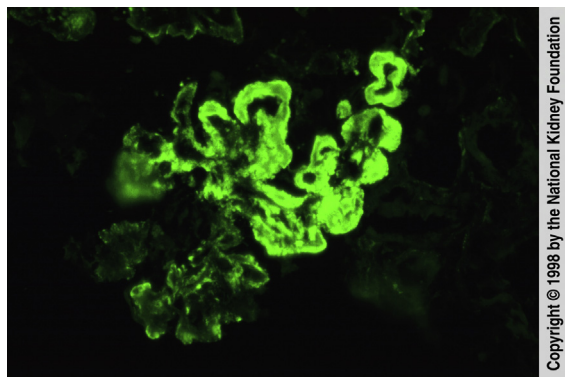


Fig 9. High-power view of diffuse proliferative lupus nephritis (WHO Class IV) shows large, sausage-shaped capillary wall deposits with smooth outer contours, indicative of subendothelial deposits, and scattered small granular capillary wall deposits, likely subepithelial deposits. Occasional mesangial deposits are also present. In lupus nephritis, staining of deposits with antisera to all three immunoglobulins, IgG, IgM and IgA, is typical (immunofluorescence with antibody to IgG; original magnification $\times 400$).

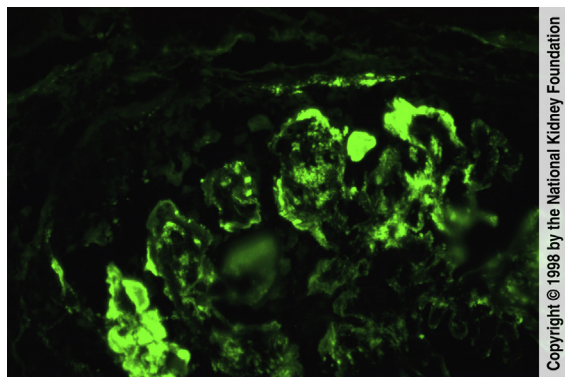
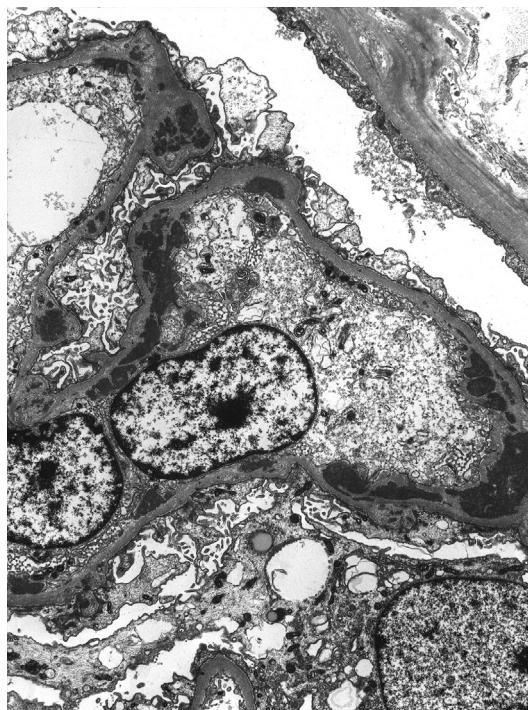
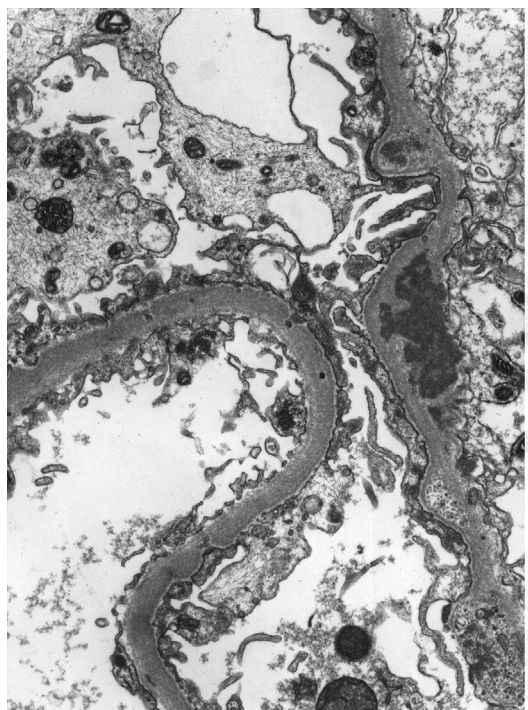


Fig 10. Diffuse proliferative lupus nephritis (WHO Class IV) with deposits that also stain for IgA (same case as in Figure 9). Large, sausage-shaped capillary wall deposits with smooth outer contours, indicating subendothelial location, are present in addition to scattered granular capillary wall and mesangial deposits. The segmental nature of deposits is apparent. A similar staining pattern, but in less intensity, was present for IgM, C3, and much weaker for C4 (immunofluorescence with antisera to IgA; original magnification $\times 400$).



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Fig 11. The endocapillary proliferation and subendothelial deposits characteristic of proliferative lupus nephritis are illustrated in this electron micrograph. Occasional small subepithelial, as well as mesangial, deposits are present. The subendothelial deposits range from small and sliver-shaped to very large, with new basement membrane formation underlying the deposits. The capillary loops are segmentally occluded by proliferating endothelial and mesangial cells (transmission electron micrograph; original magnification $\times 8,000$).



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Fig 12. High-powered electron micrograph illustrates the characteristic deposits of proliferative lupus nephritis. On the right, a large subendothelial deposit and several smaller subendothelial deposits are present. In addition, intramembranous and subepithelial deposits with a prominent spherule component (bottom right) are present. This type of organized substructure may be present in lupus, and deposits occasionally also show fingerprinting-type substructure. In endothelial cells on the left, prominent reticular aggregates, also known as tubuloreticular arrays, are present. These organized arrays of cytoplasmic membranes are associated with high levels of alpha-interferon, and are particularly prominent in endothelial cells throughout the body in patients with SLE or HIV. Foot processes are almost diffusely effaced, and endothelial cells show moderate proliferation, partially occluding the capillary lumen on the right (transmission electron microscopy; original magnification $\times 20,250$).